Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	White Cloud
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4 and SDG 8
Project / Solution Name	LumiPath
Name (Leader)	Elson Lim Zhong Qin
Email & Phone (Leader)	elsonlimzhongqin.lzq@gmail.com; +60127761455
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Rapid advancements in Artificial Intelligence (AI) and Industry 4.0 automation reduce labor demand in routine technical roles, causing workforce displacement across industrial sectors in Malaysia. According to the WEF Future of Jobs Report 2025, automation poses a 28.5% substitution risk to more than 2,800 granular skills. Displaced workers face skill depreciation, wage stagnation, and a severe lack of transparent upskilling pathways. Our project provides structured, data-backed career transition pathways for workers adapting to shifting industrial demands.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

We introduce LumiPath under the slogan “We will light the path to your future” to navigate career transitions and skill upgrades. LumiPath extracts unstructured resumes into standardized worker profiles. It integrates O*NET 28.1 data to assess emerging and declining skills, and executes deterministic mathematical matching (0.7 skill overlap + 0.3 tenure weight) to rank suitable target positions. The platform generates a 3-stage learning pathway comprising foundational gap-filling, advanced skill acquisition, and statutory certifications linked to HRD Corp courses. LumiPath models transition risks across two scenarios: steady evolution (3–5 years) and rapid disruption (1–2 years), indicating automation timelines to guide worker decisions. Technically, LumiPath connects Google Gemini 2.5 Flash API, RunPod Serverless containers, and a lightweight Flask architecture hosted on Replit.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Presentation Layer: Built with HTML/CSS/JS featuring a responsive, fluid state-flow interface styled with a midnight navy and amber gold color palette.

Core Calculation Layer: Developed on Python 3.10+, featuring RoleMatcherService (a transparent, explainable algorithm utilizing a 0.7 skill overlap and 0.3 tenure factor) and PathwayPlannerService (for automated 3-stage course and micro-credential retrieval).

AI & Reliability: Powered by Google Gemini 2.5 Flash, reinforced by an 8.0-second network hard-timeout circuit breaker and a pre-stored high-fidelity local snapshot safety fallback mechanism.

Authoritative Data Baselines: Anchored in the US Department of Labor O*NET 28.1, Malaysia TalentCorp Critical Occupations List (COL), NIOSH Malaysia statutory syllabus, and UN A/RES/70/1.

System block diagram:



Official SDC Sponsor Usage: Yes, our team used the RunPod Serverless for scalable GPU and CPU compute and inference endpoint. In addition, Replit was used as an online prototype runtime environment.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

LumiPath implements an editable human-in-the-loop workflow: users review and freely modify extracted profile fields and skill tags prior to computation. To address automation uncertainty, the interface visualizes dual disruption timelines (steady evolution vs rapid disruption), giving workers transparent time horizons. The prototype deploys on a native HTML/CSS/JS frontend served by Flask to eliminate multi-step navigation. Backend batch inference executes via RunPod Serverless containers, scaling dynamically under demand and idling at zero cost.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4 "Increase the number for youth and adults that have relevant skills for employment, entrepreneurship and decent jobs." Target 8.2 "Achieve higher level of economic productivity diversification, technological upgrading and innovation."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	28% of the more than 2800 granular skills face moderate capacity of replacement. According to Labour Market Review, Second Quarter 2026 by DOSM, the skill-related underemployment rate stood at 35.2%.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	• Alignment rate of reshaped roles with National Critical Occupations List (COL): **100%** • Target role skill gap resolution coverage: **100%** • Employer training levy claimability coverage (HRD Corp Claimable) for advanced certifications: 100% by Phase 3

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Scale to 5+ industrial fields by final prototype presentation.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python 3.11	code	PSF License
Flask 3.0.0	code	BSD-3-Clause
flask-cors 4.0.0	code	MIT
pypdf 3.17.0	code	BSD-3-Clause
python-docx 1.1.0	code	MIT
requests 2.28.0	code	Apache-2.0
runpod 1.7.0	code	MIT
Google Gemini 2.5 Flash	model	Google GenAI Terms of Service
O*NET 28.1 Database	dataset	CC BY 4.0
Hays Asia Salary Guide 2025	dataset	Public Citation / Fair Use

TalentCorp Critical Occupations List	dataset	Open Government Data
Curated HRD Corp Course Catalog	dataset	Apache-2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

10/23/2026